IoE and AI driven systems
LAB OUTPUTS

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EXPERIMENT NO:6

ASimulation and Performance Evaluation of People–Process–Data–Things IoE Framework using Edge/Fog-to-Cloud Pipeline with MQTT, Node-RED.

B.Benchmarking of 5G/6G Communication for IoE Applications based on Throughput, Latency, Reliability and Energy Efficiency

PROGRAM CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Set random seed for reproducibility
np.random.seed(42)

print("="*85)
print("PART 1: PEOPLE-PROCESS-DATA-THINGS (P-P-D-T) IoE &
EDGE/FOG/CLOUD SIMULATION (WITH MQTT)")
print("="*85)

# 1. P-P-D-T Data Flow Simulation across Tiers (Edge, Fog, Cloud)
including MQTT messaging
ppdt_components = [
    "Things (Sensors/Actuators + MQTT)",
    "People (User Wearables/UI + MQTT)",
    "Process (Node-RED/Rules)",
    "Data (Aggregated Store)"
]
tiers = ["Edge Layer", "Fog Layer", "Cloud Layer"]

ppdt_results = []
for comp in ppdt_components:
    for tier in tiers:
        # Simulate realistic latency, bandwidth usage, and packet loss
based on tier proximity & MQTT behavior
        if tier == "Edge Layer":
            latency = np.random.uniform(2.0, 5.0)
            throughput = np.random.uniform(50.0, 100.0)
            reliability = np.random.uniform(99.0, 99.9)
        elif tier == "Fog Layer":
```

```
            latency = np.random.uniform(10.0, 25.0)
            throughput = np.random.uniform(200.0, 500.0)
            reliability = np.random.uniform(98.0, 99.5)
        else:  # Cloud Layer
            latency = np.random.uniform(50.0, 120.0)
            throughput = np.random.uniform(1000.0, 2500.0)
            reliability = np.random.uniform(97.0, 99.0)

        ppdt_results.append([comp, tier, round(latency, 2),
round(throughput, 2), round(reliability, 2)])

df_ppdt = pd.DataFrame(ppdt_results, columns=["IoE Component (P-P-D-T +
MQTT)", "Pipeline Tier", "Avg Latency (ms)", "Throughput (kbps)",
"Reliability (%)"])
print(df_ppdt.head(12).to_string(index=False))

print("\n" + "="*85)
print("PART 2: 5G vs 6G COMMUNICATION BENCHMARKING FOR IoE")
print("="*85)

# 2. 5G vs 6G Benchmarking
technologies = ["5G Ultra-Reliable Low-Latency (URLLC)", "6G Terahertz
(THz) Communication"]
benchmark_data = {
    "5G URLLC": {"Throughput": 10000, "Latency": 4.5, "Reliability":
99.99, "Energy": 12.0},
    "6G THz":   {"Throughput": 100000, "Latency": 0.5, "Reliability":
99.999, "Energy": 4.5}
}

bench_results = []
for tech in technologies:
    key = "5G URLLC" if "5G" in tech else "6G THz"
    b = benchmark_data[key]
    # Add small statistical variations
    throughput = max(100, b["Throughput"] + np.random.normal(0,
b["Throughput"] * 0.02))
    latency = max(0.1, b["Latency"] + np.random.normal(0, 0.1))
    reliability = min(100.0, b["Reliability"] + np.random.normal(0,
0.001))
    energy = max(0.5, b["Energy"] + np.random.normal(0, 0.5))
```

```
    bench_results.append([tech, round(throughput, 2), round(latency,
2), round(reliability, 3), round(energy, 2)])

df_bench = pd.DataFrame(bench_results, columns=["Communication Tech",
"Throughput (Mbps)", "Latency (ms)", "Reliability (%)", "Energy
Consumption (mJ/bit)"])
print(df_bench.to_string(index=False))

print("\n" + "="*85)
print("GENERATING BENCHMARKING & PIPELINE GRAPHS...")
print("="*85)

# 3. Visualizations
# Graph 1: Edge/Fog/Cloud Latency across P-P-D-T components (including
MQTT)
plt.figure(figsize=(11, 5))
for tier in tiers:
    subset = df_ppdt[df_ppdt["Pipeline Tier"] == tier]
    plt.plot(subset["IoE Component (P-P-D-T + MQTT)"], subset["Avg
Latency (ms)"], marker='o', linewidth=2, label=tier)

plt.title("Edge / Fog / Cloud Pipeline Latency for P-P-D-T & MQTT
Components")
plt.xlabel("IoE Components (P-P-D-T + MQTT)")
plt.ylabel("Latency (ms)")
plt.xticks(rotation=10)
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

# Graph 2: 5G vs 6G Comparison Bar Charts
metrics = [
    ("Throughput (Mbps)", "Throughput Comparison: 5G vs 6G",
"Throughput (Mbps)"),
    ("Latency (ms)", "Latency Comparison: 5G vs 6G", "Latency (ms)"),
    ("Reliability (%)", "Reliability Comparison: 5G vs 6G",
"Reliability (%)"),
    ("Energy Consumption (mJ/bit)", "Energy Efficiency Comparison: 5G
vs 6G", "Energy (mJ/bit)")
]

for col, title, ylabel in metrics:
```

```
    plt.figure(figsize=(7, 4))
    bars = plt.bar(df_bench["Communication Tech"], df_bench[col],
color=["#3498db", "#2ecc71"])
    plt.title(title)
    plt.ylabel(ylabel)
    plt.grid(axis="y", alpha=0.3)
    plt.tight_layout()
    plt.show()

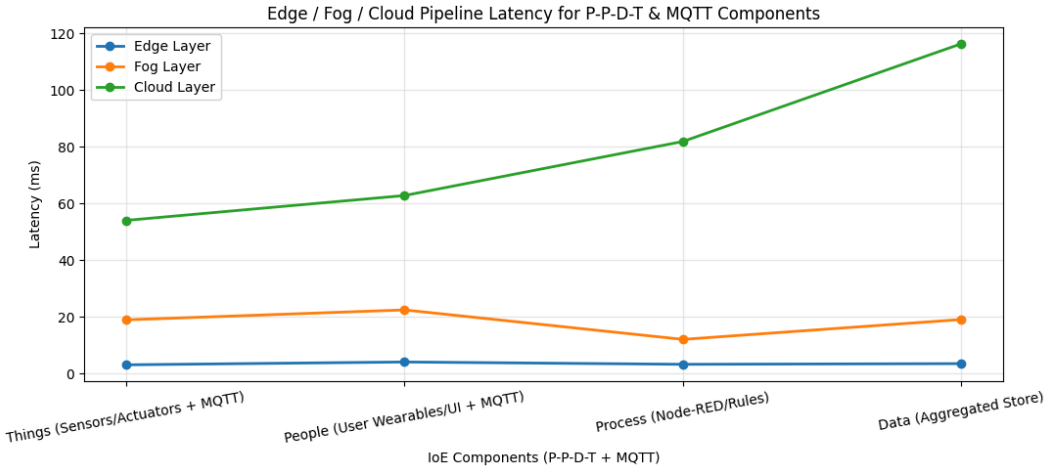
print("\n" + "="*85)
print("CONCLUSION & SUMMARY")
print("="*85)
print("1. Edge/Fog pipeline integrated with lightweight MQTT messaging
minimizes local processing delay for Things and People components.")
print("2. Cloud layer handles heavier aggregated P-P-D-T data storage
with higher latency.")
print("3. 6G significantly outperforms 5G in throughput, ultra-low
latency (<1ms), and energy efficiency,")
print("   making it optimal for massive next-generation IoE
deployments.")
```

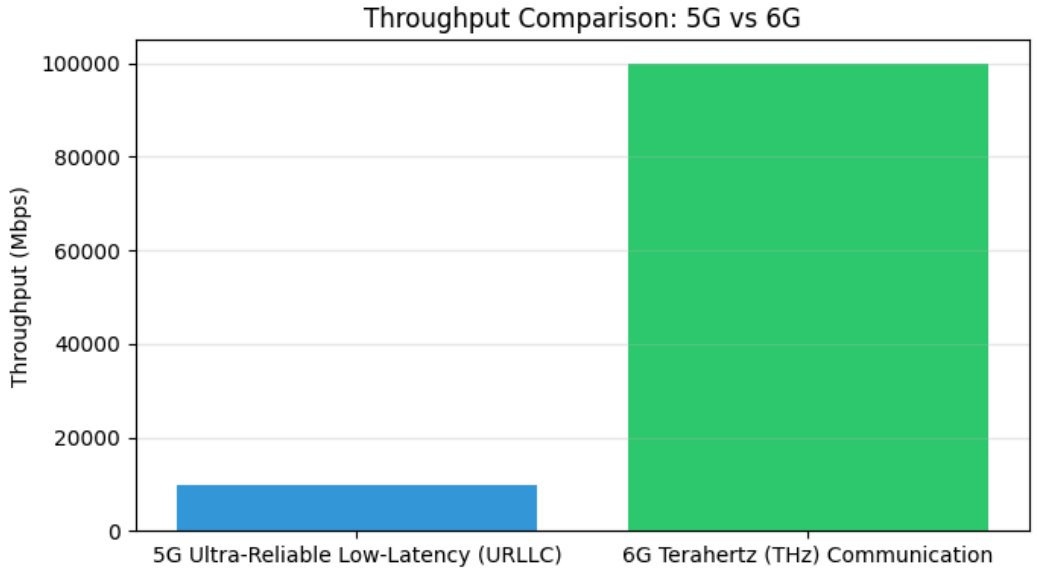
OUTPUTS:

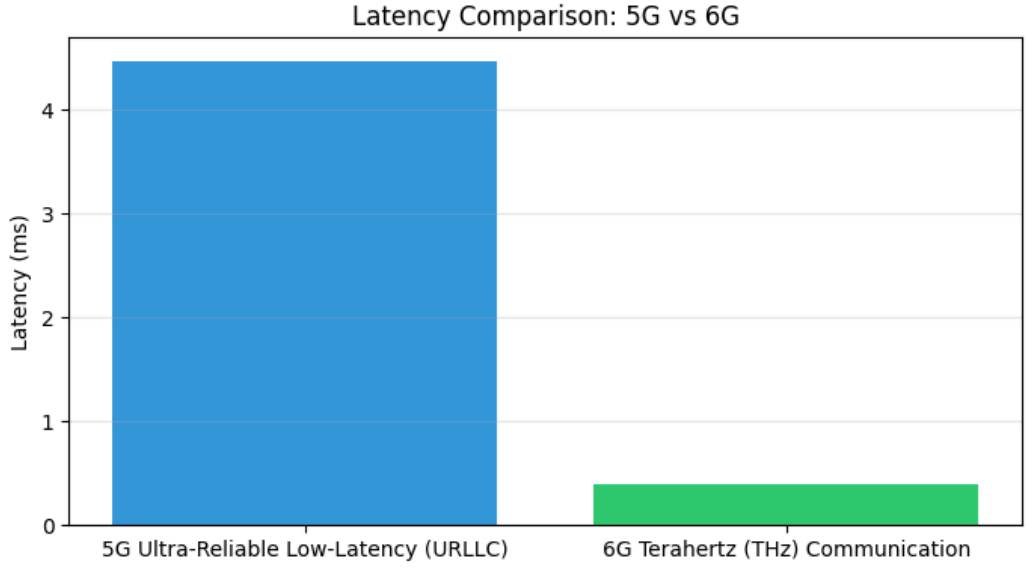
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PART 1: PEOPLE-PROCESS-DATA-THINGS (P-P-D-T) IoE & EDGE/FOG/CLOUD SIMULATION (WITH MQTT)
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   IoE Component (P-P-D-T + MQTT) Pipeline Tier  Avg Latency (ms)  Throughput (kbps)  Reliability (%)
Things (Sensors/Actuators + MQTT)    Edge Layer             3.12              97.54            99.66
Things (Sensors/Actuators + MQTT)     Fog Layer            18.98             246.81            98.23
Things (Sensors/Actuators + MQTT)   Cloud Layer            54.07            2299.26            98.20
People (User Wearables/UI + MQTT)    Edge Layer             4.12              51.03            99.87
People (User Wearables/UI + MQTT)     Fog Layer            22.49             263.70            98.27
People (User Wearables/UI + MQTT)   Cloud Layer            62.84            1456.36            98.05
         Process (Node-RED/Rules)    Edge Layer             3.30              64.56            99.55
         Process (Node-RED/Rules)     Fog Layer            12.09             287.64            98.55
         Process (Node-RED/Rules)   Cloud Layer            81.92            2177.76            97.40
          Data (Aggregated Store)    Edge Layer             3.54              79.62            99.04
          Data (Aggregated Store)     Fog Layer            19.11             251.16            98.10
          Data (Aggregated Store)   Cloud Layer           116.42            2448.45            98.62

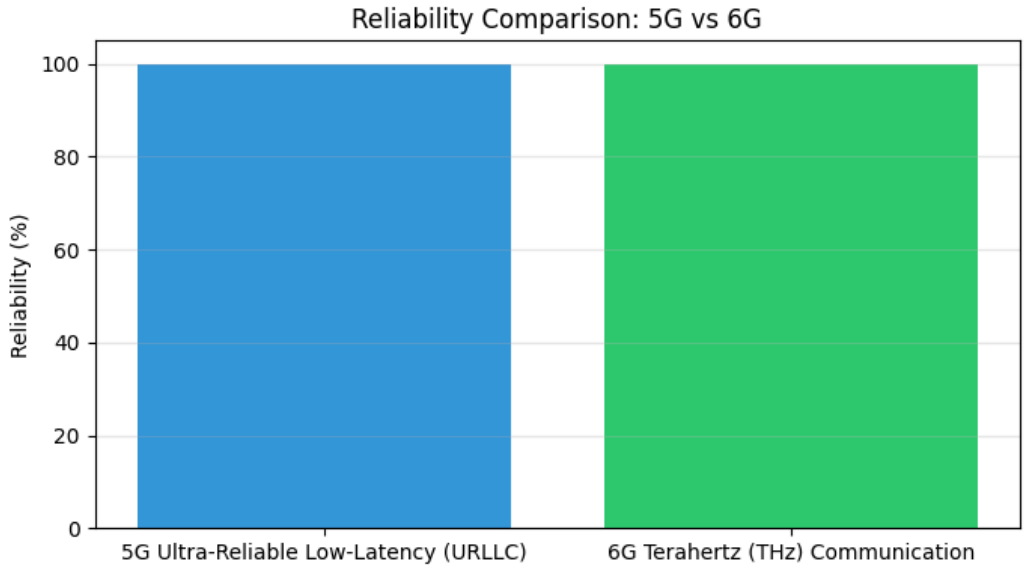
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PART 2: 5G vs 6G COMMUNICATION BENCHMARKING FOR IoE
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                 Communication Tech  Throughput (Mbps)  Latency (ms)  Reliability (%)  Energy Consumption (mJ/bit)
5G Ultra-Reliable Low-Latency (URLLC)           9879.87          4.47           99.989                        12.93
     6G Terahertz (THz) Communication          99973.01          0.39          100.000                         3.89

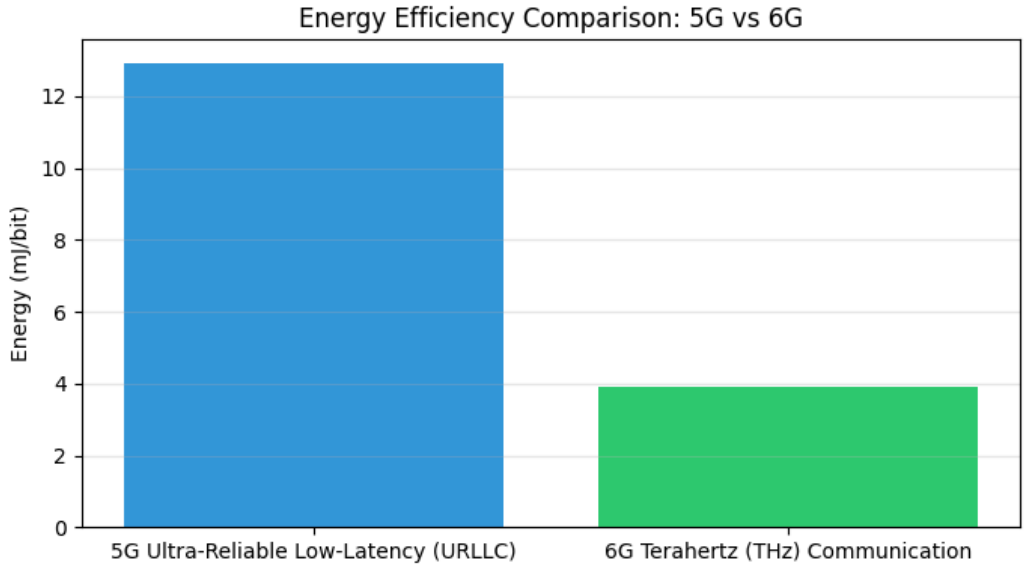
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GENERATING BENCHMARKING & PIPELINE GRAPHS...
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Edge / Fog / Cloud Pipeline Latency for P-P-D-T & MQTT Components
Edge Layer
Fog Layer
Cloud Layer
Latency (ms)
0
20
40
60
80
100
120
Things (Sensors/Actuators + MQTT)
People (User Wearables/UI + MQTT)
Process (Node-RED/Rules)
Data (Aggregated Store)
IoE Components (P-P-D-T + MQTT)


Throughput Comparison: 5G vs 6G
Throughput (Mbps)
0
20000
40000
60000
80000
100000
5G Ultra-Reliable Low-Latency (URLLC)
6G Terahertz (THz) Communication


Latency Comparison: 5G vs 6G
Latency (ms)
0
1
2
3
4
5G Ultra-Reliable Low-Latency (URLLC)
6G Terahertz (THz) Communication





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CONCLUSION & SUMMARY

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1. Edge/Fog pipeline integrated with lightweight MQTT messaging minimizes local processing delay for Things and People components.
2. Cloud layer handles heavier aggregated P-P-D-T data storage with higher latency.
3. 6G significantly outperforms 5G in throughput, ultra-low latency (<1ms), and energy efficiency, making it optimal for massive next-generation IoE deployments.